Prodigy Anesthesia Quick Review

Barbiturates

What are the chemical origins of barbiturates?	Barbiturates are derived from barbituric acid. (Barbituric acid is formed from the condensation of urea and malonic acid.) *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 127.*
To what receptor do barbiturates bind with to exert their clinical action and where is this receptor located?	Barbiturates bind with the GABA-A receptor found in neurons in the central nervous system. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 378.*
What chemically defines the oxybarbiturates? What are some examples of oxybarbiturates?	Oxybarbiturates are known as true barbiturates and have an oxygen at the second carbon atom. Examples of oxybarbiturates include pentobarbital, methohexital, secobarbital, and phenobarbital. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 176.* *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 103.*
What chemically defines the thiobarbiturates? What is an example of this class of drug?	The thiobarbiturates replace the oxygen at the second carbon atom with a sulfur atom. Thiopental is an example of a thiobarbiturate. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 176.*
Are barbiturates weak acids or weak bases?	They are weak acids and are prepared as sodium salts. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 127-130.*
What chemically defines the methylated oxybarbiturates? What is an example of this class of drug?	The methylated oxybarbiturates retain the oxygen at the second carbon atom, but have a methyl group on the first carbon atom. Methohexital is a methylated oxybarbiturate. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 127.*
How do barbiturates affect cerebral blood flow and metabolism?	Barbiturates reduce cerebral metabolic rate, and reduce cerebral blood flow (by INCREASING cerebrovascular resistance). *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 583.*

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How do barbiturates produce an inverse steal phenomenon in the brain?	Barbiturates produce vasoconstriction only in normal cerebral tissue resulting in preferential blood flow to ischemic areas while simultaneously reducing total cerebral blood flow. This is referred to as the Robin Hood effect or inverse steal phenomenon. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 583.*
What is the pKa of thiopental?	The pKa of thiopental is 7.6. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 129-130.*
Does thiopental provide analgesia?	No. In fact, in low doses, thiopental has proven to produce hyperalgesia. *Ouellette RG, Joyce JA.Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones and Bartlett Learning; 2011: 43.*
Can thiopental be reconstituted with Lactated Ringer's solution? Why or why not?	Barbiturates must be reconstituted with sterile saline or sterile water. Even though barbiturates are weak acids, they are prepared with anhydrous sodium carbonate which makes the powder alkaline. If Lactated Ringer's solution or any other acidic solution is used, the thiopental will precipitate out of the solution. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 103.*
Can thiopental cross the placental barrier?	Yes. Thiopental is extremely lipid-soluble and can easily cross the blood-brain barrier or the placental barrier. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 130, 136.*
What accounts for the short duration of action of thiopental? What part does metabolism play?	The short duration of action of thiopental is attributed to the rapid redistribution away from the brain and back into the systemic circulation. Metabolism does not play a significant role in the duration of action of a single dose of thiopental. *Ouellette RG, Joyce JA.Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones and Bartlett Learning; 2011: 43.*
What are the cardiac effects of an induction dose of thiopental and what is responsible for these effects?	Thiopental produces dose-dependent decreases in stroke volume, cardiac output, and arterial blood pressure. This is mainly due to increased venous capacitance and myocardial depression. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 694-695.*

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Thiopental does not undergo significant elimination via the kidneys. Why is this?	Only about 1% of thiopental is eliminated unchanged by the kidneys. The primary reason for this is because thiopental is 80% protein-bound and the protein-drug complex is not easily filtered at the glomerulus. *Ouellette RG, Joyce JA.Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones and Bartlett Learning; 2011: 43.*
How would patient pH affect the clinical action of thiopental? Why?	Thiopental is an acidic compound. The more acidotic the patient becomes, the more nonionized the drug becomes. As a result, more of the drug is in the lipid-soluble state that can cross the blood-brain barrier and exert a clinical effect. In short, acidosis increases the effect of thiopental. Alkalosis decreases the effect of thiopental. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 176-177.*
To what degree is thiopental protein-bound?	Thiopental is 80% protein-bound. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 177.*
How long can reconstituted thiopental sit at room temperature and still be used for injection?	Reconstituted thiopental is stable and sterile at room temperature for at least 6 days. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 127.*
What are the side effects of methohexital administration?	It can produce spontaneous muscle movement, myoclonus, hiccoughs, and seizures. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 132.*
How long is reconstituted methohexital stable?	6 weeks. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 127.*
How do the distribution kinetics and metabolism of thiopental and methohexital compare?	The distribution kinetics of thiopental and methohexital are quite similar. The hepatic extraction ratio for methohexital, however, is three times that of thiopental. *Ouellette RG, Joyce JA.Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones and Bartlett Learning; 2011: 42-43.*

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